

JONATHAN ALEXANDER YATES

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EDUCATION

Purdue University | *B.S. in Aerospace Engineering*

West Lafayette, IN | *Expected May 2029*

Minor in Electrical & Computer Engineering | GPA: **3.78 / 4.00** | Honors: Dean's List & Semester Honors (2025, 2026)

Relevant Coursework: Aeromechanics, Linear Circuit Analysis, Multivariable Calculus, Ordinary Differential Equations, Transform Methods in Systems Engineering, Mechanics of Materials

TECHNICAL SKILLS

CAD & Simulation: Autodesk Fusion 360 (Certified), Siemens NX, FEA & Vibration Analysis, GD&T, 3D Printing / Slicing

Controls & Automation: Rockwell Studio 5000, RSLogix 500, FactoryTalk Optix & View, GE iFIX, MATLAB, Python, C

Electronics & Hardware: Industrial Ethernet/IP, IP Encoders & Decoders, Sensor Wiring, Soldering, Avionics Packaging

Machining & Fabrication: CNC Mill (CAM), Manual Lathe, GMAW/SMAW Welding (CTE Certified), Band Saws

ENGINEERING PROJECTS & STUDENT TEAMS

Purdue Space Program Active Controls | *Structures Team*

Purdue University | *Aug 2025 - Present*

Lead

- **Structures Leadership & Coordination:** Lead the structures sub-team for the TOAD actively stabilized lander rocket; coordinate airframe load path requirements, CAD modeling reviews, and fabrication timelines across sub-teams.
- **TOAD Lander Rocket:** Designed custom electronics mounting brackets in Siemens NX and ran FEA vibration simulations to make sure flight computers and sensors survive landing impact and thruster loads.
- **ASTRA Avionics Testbed:** Teamed up with avionics members to design chassis frame parts and physically build a rapid-prototyping benchtop rig to test and debug the lander's flight computer.
- **Hardware Packaging & Routing:** Helped route wire harnesses and securely package flight avionics inside the chassis so sensor connections stay clean and reliable.
- **Machining & CAM:** Generated CAM toolpaths in Fusion 360 and machined airframe brackets and motor mounts using CNC mills, manual lathes, and shop welding equipment.

Solar Oven Engineering Challenge | *1st Place Team Winner*

University of Arizona | *2024*

- Designed, modeled, and tested a parabolic solar thermal oven with a student team, earning 1st place for structural design and heat concentration.

WORK EXPERIENCE & INTERNSHIPS

NSG Pilkington | *Electrical & Systems Engineering Intern*

Laurinburg, NC | *Summer 2026*

- **Hands-On Training Board:** Built and wired a physical PLC/HMI training station from scratch—connected buttons, selector switches, indicator lights, and touchscreens to Rockwell PLCs so technicians could simulate real line operations.
- **IP Video & Signal Networking:** Set up and networked IP encoders and decoders over plant Ethernet/IP to send low-latency video feeds and equipment status data directly to operator stations.
- **PLC Logic & Troubleshooting:** Wrote, tested, and troubleshoot ladder logic routines and safety interlocks using RSLogix 500 and Studio 5000.
- **HMI Dashboards:** Designed clear touch-screen interfaces and alarm monitoring screens for plant operators using FactoryTalk Optix and GE iFIX.
- **Network Setup:** Assisted in configuring IP subnets and dedicated servers for automated machine-vision cameras and remote I/O blocks.

Smithsonian MMT Observatory | *Engineering Shadow*

Mount Hopkins, AZ | *Summer 2024*

- Shadowed lead engineers during maintenance and optical alignment on large telescope assemblies; observed diagnostic checkouts and servo tuning on high-precision motion systems.

Thunderbolt Welding | *Fabricator & Welder*

Midland, NC | *2022 - 2024*

- Fabricated custom steel fixtures and trailers from blueprints using band saws, ironworkers, and welding rigs following OSHA safety practices.

LEADERSHIP & CERTIFICATIONS

Purdue Scuba Club | *Vice President*

Purdue University | *Aug 2025 - Present*

- Help run club operations, inspect technical dive gear and regulators, and coordinate trip logistics and safety planning for student dive trips.

Certifications: Autodesk Certified User (Fusion 360 CAD/CAM) | OSHA-10 General Industry | CTE Welding & Fabrication | SSI Advanced Open Water Diver